

PROBLEMS, EXERCISES, AND QUESTIONS

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Preface

This document contains the essential theory, conceptual questions, exercises, and problems in Center of Mass and Moment of Inertia. Each chapter begins with basic definitions on how to calculate the corresponding quantities, proceeds to display useful tools when doing calculations, and ends with the necessary theorems.

The problem sets cover all of the basic results that you will encounter in an AP Physics 1 / C course along with a little extra. If plan to solve the problems using calculus, you can skip the questions and exercises and purely read through the theory before proceeding onto the problem set. If you plan to solve all the problems using algebra, I highly recommend you work through the questions and exercises as key ideas are illustrated in each that are not re-explained in the solutions to the problem sets.

The calculus solutions are standard, done in either cartesian or polar coordinates. The algebra solutions can be summed up as compilation of the classic scaling arguments, lots of superposition, and an assortment of other tricks and techniques. The majority of them require more thinking than math, meaning they require a deeper understanding of what you're actually doing.

If you do everything with algebra, you'll find that deriving formulas takes very little time and memorizing them takes little effort. As always, if you see anyone trying to integrate a circle in cartesian coordinates or struggling to remember the moment of inertia of a hoop, please share the document with them.

Solutions to the questions, exercises, and algebra solutions for the problems are located in the solutions manual. Calculus solutions are located at the end of the each chapter. As there are no example problems, don't hesitate too much to look at the solution the first time you encounter a new type of problem. Of course, after that give each problem a good amount of thought.

Finally, although I have re-read the entire document, there are bound to be typos. If you spot an error or find anything that doesn't make sense, I'd really appreciate it if you let me by email: neerkuchlous@gmail.com. Thanks, and enjoy!

Neer Kuchlous

Chapter 1

Center of Mass

1.1 Definitions

Point

Consider a point P in space relative to an origin O .

$$\boldsymbol{\psi} = m\mathbf{r} \quad (1.1)$$

$\boldsymbol{\psi}$ represents the position-mass vector of P ; m represents the mass of P ; $\mathbf{r}$ represents the position vector of P relative to O . (Fig. 1.1)

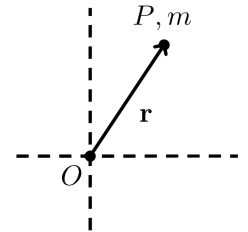


Figure 1.1

Discrete Definition

Consider a collection of N points in space with an arbitrary origin O .

$$\boldsymbol{\psi} = \sum_{k=1}^N m_k \mathbf{r}_k \quad (1.2)$$

$\boldsymbol{\psi}$ represents the total position-mass of the N points; m_k represents the mass of the k th point; $\mathbf{r}_k$ represents the position vector of the k th point with respect to O .

Continuous Definition

Consider an object S in space with an arbitrary origin O .

$$\boldsymbol{\psi} = \int_S \mathbf{r} dm \quad (1.3)$$

$\boldsymbol{\psi}$ represents the total position-mass of S ; dm represents a differential mass of S ; $\mathbf{r}$ represents the position vector of a dm with respect to O . (Fig. 1.2)

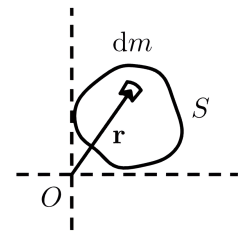


Figure 1.2

Superposition

The total position-mass of a system S equals the sum of its parts, which can be arbitrary grouped and chosen.

Center of Mass

We can represent an enclosed system S with a point P of equivalent mass M such that the P and S have equal total position-mass $\boldsymbol{\psi}$. The position of the point $\mathbf{r}_{CM}$ is known as the center of mass of S .

Questions

Q1.1) Show how the discrete definition for the position-mass of an object follows from the definition of the position-mass of a point.

Q1.2) Show how the continuous definition for the position-mass of an object follows from the discrete definition of the position-mass of a system.

Q1.3) Make sure you intuitively understand how superposition follows from the associative property of addition when adding the position-mass of multiple points.

Q1.4) Derive the center of mass formula for a system with discrete mass distribution and a system with continuous mass distribution.

Q1.5) Derive the general formula for the center of mass of two points of mass m_1 and m_2 located at x_1 and x_2 respectively.

Q1.6) Explain how the principle of superposition suggests that the center of mass of an object can be found by finding the center of mass of its parts and taking the weighted average of the centers.

Q1.7) Consider an object such that the mass of every point is scaled by a factor k . Show that the location of the center of mass does not change.

Exercises

E1.8) Find the center of mass of three points with masses m , $2m$, and $3m$ that are located at $(-2d, d)$, $(-d, 3d)$, and $(3d, 0)$ on the xy -plane (Fig 1.3).

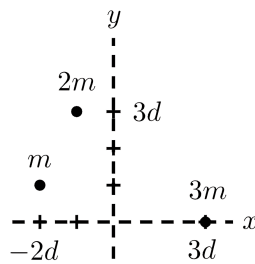


Figure 1.3

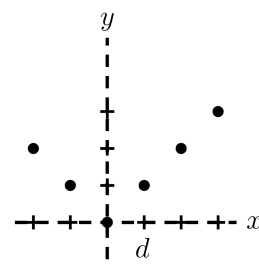


Figure 1.4

E1.9) Find the center of mass of the 6 points, each with mass m , that form a V shape figure as shown (Fig 1.4). Adjacent lines are spaced a distance d apart.

E1.10) A cube with side length ℓ has six faces. One is cut out such that the remaining faces have a total mass $5m$. Find the center of mass of the resulting shape.

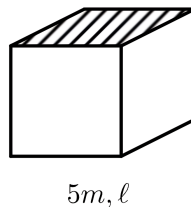


Figure 1.5

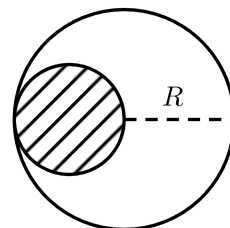


Figure 1.6

E1.11) A circle has a radius R . A smaller circle with radius $R/4$ is cut out from the larger circle such that both circles share a point of tangency. Find the center of mass of the resulting shape.

E1.12) From $x = d$ to $x = Nd$, N equally sized points lie on the x -axis. Find the center of mass relative to the origin O .

1.2 Tools

Symmetry

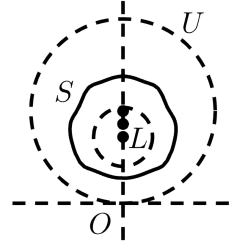
Given that an object has cylindrical symmetry, the center of mass must lie upon it.

Estimation

Consider an object S which is enclosed by a smaller object L and a larger object U such that L , S , and U share an axis of cylindrical symmetry.

$$\mathbf{r}_L \leq \mathbf{r}_S \leq \mathbf{r}_U \quad (1.4)$$

$\mathbf{r}_L$, $\mathbf{r}_S$, and $\mathbf{r}_U$ are the center of masses of L , S , and U respectively. (Fig. 1.7)



Scaling Arguments

Consider an object S that is scaled by a factor k about the origin O to get S_k .

$$\psi_k = \psi_0 k^{r+1} \quad (1.5)$$

ψ_k is the total position mass of S_k ; ψ_0 is the total position mass of S ; k represents the scale factor; r represents an integer constant that is dependent on the geometry of the object. (Fig. 1.8)

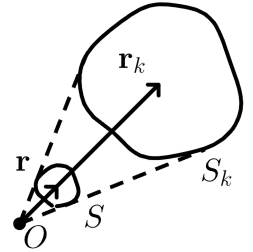


Figure 1.8

Limiting Arguments

Consider an object S_1 and a smaller scaled down version S_2 . Superimpose S_2 upon S_1 such that S_3 is formed by their difference.

$$\lim_{S_2 \rightarrow S_1} \psi_3 = \lim_{S_2 \rightarrow S_1} \psi_1 - \psi_2 \quad (1.6)$$

ψ_1 , ψ_2 , and ψ_3 represent the total position-mass of S_1 , S_2 , and S_3 respectively. (Fig. 1.9)

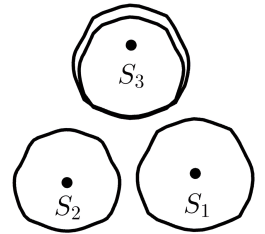


Figure 1.9

Questions

Q1.13) Consider a hollow cylinder with the center of mass located a distance d away from the cylindrical axis. Rotate the cylinder about some angle θ and show that $d = 0$ is the only viable solution.

Q1.14) Utilizing symmetry and center of mass by parts, show that any object whose density varies radially only must have a center of mass that lies on the cylindrical axis.

Q1.15) Explain the importance of a shared axis of cylindrical symmetry when forming a lower L and upper U bound for an object S .

Q1.16) Suppose the center of mass of an object S is found to be $\mathbf{r}_{cm}$ with respect to the origin O . Let the new origin O' be defined such that the vector from O' to O is $\Delta \mathbf{r}$. Show that $\mathbf{r}_{cm} + \Delta \mathbf{r}$ gives the new center of mass of S relative to O' .

Q1.17) Evaluate the total position-mass of a system S relative to an origin O located at the center of mass.

Q1.18) Show that when an object is scaled by a factor k , lengths are scaled by a factor k , areas are scaled by a factor k^2 , and volumes are scaled by a factor k^3 . Generalize this to an n dimensional object.

Q1.19) Consider an n dimensional object S with nonuniform density such that the density varies according to $\rho = x^s$. Show that when S is scaled by a factor k , the mass m of S is scaled by a factor k^{n+s} .

Q1.20) Building on the previous exercise, we seek to prove Eq 1.5. Do this by evaluating the change in the position-mass ψ of a point after scaling it by a factor of k and then summing up the position-mass over all the points in a given system S .

Q1.21) Show that Eq. (1.6) follows from superposition and evaluate the limit.

Exercises

E1.22) A half-hoop has a radius R . Estimate the center of mass by bounding the half-hoop between an upper and lower bound (Fig. 1.10) .

E1.23) Estimate the center of mass of a half-circle, also with radius R . It will be useful to know the center of mass of a triangle lies $1/3$ of the distance from its base.

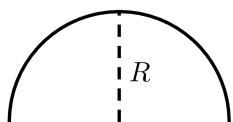


Figure 1.10

1.3 Theorems

N Points Theorem

Consider an n dimensional ($0 \leq n \leq 3$) object S with $N = n + 1$ vertices such that the object has uniform mass density.

$$\mathbf{r}_{\text{CM}} = \frac{1}{N} \sum_{k=1}^N \mathbf{r}_k \quad (1.7)$$

$\mathbf{r}_{\text{CM}}$ represents the center of mass of S ; $\mathbf{r}_k$ represents the position of the k th vertex. (Fig. 1.11)

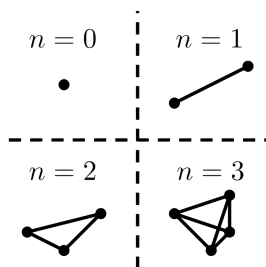


Figure 1.11

Questions

Q1.24) Evaluate Eq. 1.7 for $n = 0$ and $n = 1$. Then explain why uniform mass density is an important requisite by considering the limiting cases for a rod.

Exercises

E1.25) A line has two ends located at $x = 0$ and $x = \ell$ respectively. The density of the line varies according to $\rho = x$. Find the center of mass of the line.

E1.26) The same line now has a density that varies according to $\rho = x^2$. Find the new center of mass (think about how can you make the N -points theorem applicable).

1.4 Problems

1.1 N-Points Theorem Line

A line with uniform density has a length ℓ . Show that the center of mass lies halfway in between.

1.2 N-Points Theorem Triangle

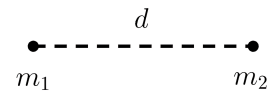
A triangle has uniform density. Prove that the center of mass lies at the average of its vertices.

1.3 N-Points Theorem Tetrahedron

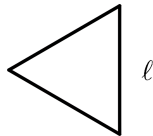
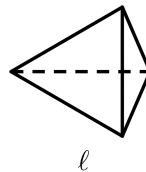
A tetrahedron has uniform density. Show that the center of mass lies at the average of its vertices.

1.4 Two Points

Two points with mass m_1 and m_2 are located a distance d apart. Find the distance of both points relative to the center of mass (Fig. 1.12) .

**Figure 1.12****1.5 Equilateral Triangle**

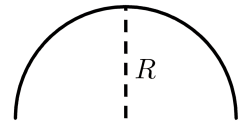
An equilateral triangle has a side length ℓ . Find the center of mass relative to one corner (Fig. 1.13).

**Figure 1.13****Figure 1.14****1.6 Regular Tetrahedron**

A regular tetrahedron has a side length ℓ . Find the center of mass relative to one corner (Fig. 1.14).

1.7 Half Hoop

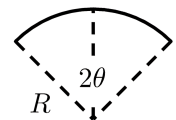
A half-hoop has a radius R . Find the center of mass relative to bottom of the hoop (Fig. 1.15) .

**Figure 1.15****1.8 Half Circle**

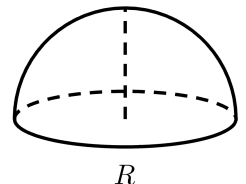
A half-circle has a radius R . Find the center of mass relative to the bottom of the circle.

1.9 Arc

An arc of a circle with radius R subtends an angle 2θ . Find the center of mass of the arc relative to the center of the circle (Fig. 1.16) .

**Figure 1.16****1.10 Hollow Hemisphere**

A hollow hemisphere has a radius R . Find the center of mass relative to the bottom, missing face (Fig. 1.17) .

**Figure 1.17****1.11 Solid Hemisphere**

A solid hemisphere has a radius R . Find the center of mass relative to the bottom face.

1.12 Trapezoid

An isosceles-trapezoid has bases a and b alongside a height h . Find the center of mass of the trapezoid relative to the larger base (Fig. 1.18).

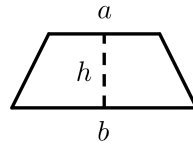


Figure 1.18

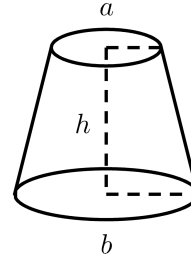


Figure 1.19

1.13 Conical Frustum

A conical frustum has radii a and b alongside a height h . Find the center of mass of the frustum relative to the larger base (Fig. 1.19).

1.5 Calculus Solutions to Problems

1.1 N-Points theorem Line

Let the rod have a mass m resulting in a linear density $\lambda = m/\ell$. We now can find the mass and position of a slice (Fig. 1.20) .

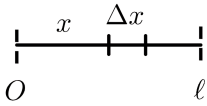


Figure 1.20

$$\Delta m = \lambda \Delta x$$

$$r(x) = x$$

We can then apply the continuous center of mass definition.

$$mr_{cm} = \int_0^\ell \lambda x dx = \frac{m}{\ell} * \frac{1}{2} x^2 \Big|_0^\ell$$

We can then solve for the center of mass to find $r_{cm} = \ell/2$

1.2 N-Points Theorem Triangle

For any triangle, we can set our coordinate system so that the y-axis is parallel to the opposite edge.

Let the triangle then have a mass m , altitude a , base b resulting in an area $A = ab/2$. We find the surface density.

$$\sigma = m/A = 2m/ab$$

Using similar triangles, we can find the height of a slice (Fig. 1.21) .

$$\frac{y}{x} = \frac{b}{a}$$

We can then find the mass $\sigma \Delta A$ and position of a slice.

$$\Delta m = \sigma y \Delta x$$

$$r(x) = x$$

We then apply the continuous center of mass definition.

$$mr_{cm} = \int_0^a \sigma y x dx = \frac{2m}{a^2} \int_0^a x^2 dx = \frac{2m}{a^2} * \frac{1}{3} x^3 \Big|_0^a$$

We can then solve for the center of mass to find $r_{cm} = 2a/3$

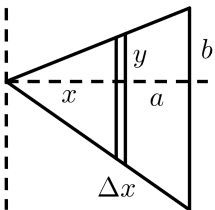


Figure 1.21

1.3 N-Points Theorem Tetrahedron

For any tetrahedron, we can choose our coordinate system such that the y -axis is parallel and the x -axis is perpendicular to the opposite face.

Let the area of any triangular face at a height be $A(y) = ky^2$. We then let the tetrahedron have a mass m , density ρ , altitude h , and a base with area $A_b = kb^2$.

We know the volume $V = A_b h/3$ and solve for density.

$$\rho = m/V = 3m/kb^2h$$

Using similar triangles, we find $y/x = b/h$. Squaring and multiplying by k , we find

$$\frac{A_y}{x^2} = \frac{kb^2}{h^2}$$

We can then find the mass $\rho \Delta V$ and position of a slice (Fig. 1.22)

$$\Delta m = \rho A_y \Delta x \quad r(x) = x$$

We then apply the continuous center of mass definition.

$$mr_{cm} = \int_0^a \rho A_y x dx = \frac{3m}{kb^2h} * \frac{kb^2}{h^2} * \frac{1}{4} x^4 \Big|_0^a$$

We can then solve for the center of mass to find $r_{cm} = 3h/4$

REMARK: We declared the area of any triangular slice equal to some unknown constant k multiplied by the length y from the x -axis to one of the edges squared. Notice how we didn't bother solving for the constant using geometry or using any of the inbuilt features of the shape in our similar triangles; rather, we defined our own variables. In the end, the constant cancels out and all the defined variables disappear. We see that what really matters is the rate of growth. This means that if the tetrahedron were instead a cone $k = \pi$, a regular tetrahedron $k = \sqrt{3}/4$, or even a square pyramid $k = 1$, the center of mass would lie at the same height.

1.4 Two Points

We let our origin O lie at the center of mass and the positions of m_1 and m_2 be x_1 and x_2 respectively. We know the total position-mass is zero (Fig. 1.23).

$$m_1 x_1 = m_2 x_2$$

We also know the individual distances add to the total distance.

$$x_1 + x_2 = d$$

Substituting and solving, we find $x_1 = \frac{m_2 d}{m_1 + m_2}$ and $x_2 = \frac{m_1 d}{m_1 + m_2}$

1.5 Equilateral Triangle

We know the center of mass must lie upon all lines of symmetry. An equilateral triangle has three, which coincide at a single point. Let's draw the altitude to one of the bases, labeling the apex as A , the foot F , and the left corner L .

Then, we draw a second line of symmetry starting from L that intersects the altitude AF at P (Fig. 1.24). By symmetry,

$$LP \cong AP$$

Let FP have a length x . You can quickly show $LF = \ell/2$ and $AF = \sqrt{3}\ell/2$. That means

$$AP = \sqrt{3}\ell/2 - x$$

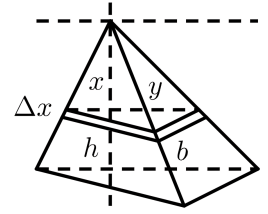


Figure 1.22

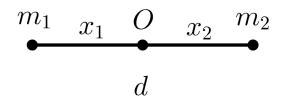


Figure 1.23

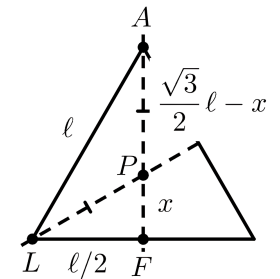


Figure 1.24

Finally, we can apply the Pythagorean theorem to triangle ΔLFP to solve for x .

$$\left(\frac{\sqrt{3}}{2}\ell - x\right)^2 = x^2 + \left(\frac{\ell}{2}\right)^2$$

Solving, we find $x = \ell/2\sqrt{3}$

REMARK: However, we can find the ratio $x/h = 1/3$, reconfirming the N -Points theorem. We can also solve for AP which becomes $\ell\left(\frac{\sqrt{3}}{2} - \frac{1}{2\sqrt{3}}\right) = \frac{\ell}{\sqrt{3}}$. This will be useful in the next problem.

1.6 Regular Tetrahedron

As before, the center of mass must lie upon all lines of symmetry. A regular tetrahedron has four, which coincide at a single point P' . We'll choose to work in the plane containing the apex A' , the foot of the altitude from the apex to the base F' , and one of the bottom vertices L' (Fig. 1.25).

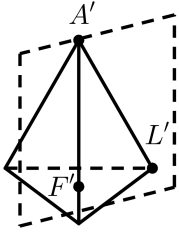


Figure 1.25

From before, we found AP which is congruent to $L'F'$ now. We also know the length of the hypotenuse (an edge on the tetrahedron). We can then solve for the height using the Pythagorean Theorem (Fig. 1.26).

$$\ell^2 = h^2 + (\ell/\sqrt{3})^2$$

We find $h = \ell\sqrt{2/3}$. As before, we can draw a second line of symmetry beginning from L' and crossing P' . By symmetry,

$$L'P' \cong A'P'$$

Let $F'P'$ have a length x meaning $A'P'$ has a length $\ell\sqrt{2/3} - x$. We can now solve for x using the Pythagorean Theorem.

$$\left(\ell\sqrt{\frac{2}{3}} - x\right)^2 = x^2 + \left(\frac{\ell}{\sqrt{3}}\right)^2$$

Solving, we find $x = \ell/2\sqrt{6}$

REMARK: As before, we can find the ratio $x/h = 1/4$, again confirming the N -Points theorem.

1.7 Half Hoop

Let the half-hoop have a mass m resulting in an angular density $\lambda = m/\pi$, allowing us to find the mass and position of a slice (Fig. 1.27).

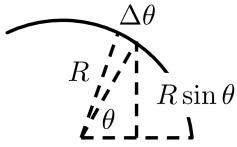


Figure 1.27

$$\Delta m = \lambda \Delta \theta$$

$$r(\theta) = R \sin \theta$$

We can then apply the continuous center of mass definition.

$$mr_{cm} = \int_0^\pi \lambda R \sin \theta d\theta = \frac{mR}{\pi} * -\cos \theta \Big|_0^\pi$$

We can then solve for the center of mass to find $r_{cm} = 2R/\pi$

1.8 Half Circle

Let the half-circle have a mass m and surface density σ . We find the area $A = \pi R^2/2$ and solve for surface density.

$$\sigma = m/A = 2m/\pi R^2$$

We can then find the mass $\sigma \Delta A$ and position of a slice (Fig. 1.28) .

$$\Delta m = \sigma r \Delta r \Delta \theta \quad r_s(\theta) = r \sin \theta$$

We can then apply the continuous center of mass definition.

$$mr_{cm} = \int_0^\pi \int_0^R \sigma r^2 \sin \theta \, dr \, d\theta = \frac{2m}{\pi R^2} * -\cos \theta \Big|_0^\pi * \frac{1}{3} r^3 \Big|_0^R$$

We can then solve for the center of mass to find $r_{cm} = 4R/3\pi$

REMARK: When evaluating the r integral, any θ term is constant and can be factored out. You can then evaluate the r integral, which then becomes constant, allowing you to factor it out and evaluate the θ integral. Otherwise, you can cut the half-circle into hoop-like slices, with mass $\Delta m = \sigma * \pi r \Delta r$ and position $r_s(r) = 2r/\pi$, utilizing the previous problems result.

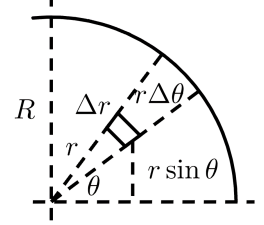


Figure 1.28

1.9 Arc

Let the arc have a mass m resulting in a linear density $\lambda = m/2\theta$, allowing us to find the mass and position of a slice (Fig. 1.29) .

$$\Delta m = \lambda \Delta \phi \quad r(\phi) = R \cos \phi$$

We can then apply the continuous center of mass definition.

$$mr_{cm} = \int_{-\theta}^{\theta} \lambda \, d\phi = \frac{m}{2\theta} * \sin \phi \Big|_{-\theta}^{\theta}$$

We can then solve for the center of mass to find $r_{cm} = R \sin \theta / \theta$

REMARK: We can evaluate the limit $\theta \rightarrow 0$ where the arc becomes a point.

$$\lim_{\theta \rightarrow 0} r_{cm} = \lim_{\theta \rightarrow 0} R * \frac{\sin \theta}{\theta} = R$$

because $\sin \theta \approx \theta$ for $\theta \ll 1$. This agrees with our intuition as the center of mass of a point simply lies on itself. Additionally, we can check $2\theta = \pi$ where the arc becomes a circle. We find $r_{cm} = 2R/\pi$ as expected.

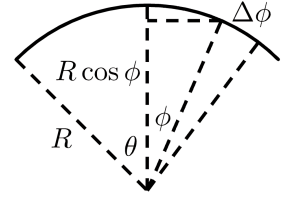


Figure 1.29

1.10 Hollow Hemisphere

Let the hemisphere have a mass m . We know the area $A = 2\pi R^2$ and can solve for the surface density $\sigma = m/2\pi R^2$.

We can then find the mass $\sigma 2\pi r \Delta s$ and vertical position of a slice (Fig. 1.30) .

$$\Delta m = 2\pi R \cos \theta * R \Delta \theta \quad r(\theta) = R \sin \theta$$

At last, we apply the continuous center of mass definition.

$$mr_{cm} = \int_0^{\pi/2} \sigma 2\pi R^3 \cos \theta \sin \theta \, d\theta$$

We can use $\sin 2\theta = 2 \sin \theta \cos \theta$ to simplify and evaluate the integral.

$$mr_{cm} = \frac{m}{2\pi R^2} * \frac{2\pi R^3}{1} \int_0^{\pi/2} \sin 2\theta \, d\theta = \frac{mR}{2} * -\frac{1}{2} \cos 2\theta \Big|_0^{\pi/2}$$

We can then solve for the center of mass to find $r_{cm} = R/2$

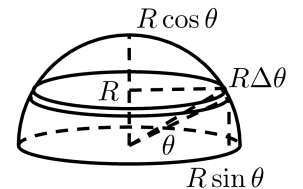


Figure 1.30

1.11 Solid Hemisphere

Let the solid hemisphere have a mass m . We know the volume $V = \frac{2}{3}\pi R^3$ and can solve for volume density $\rho = m/V$.

Using Pythagorean's theorem, we can relate the radius of a slice x , height y , and the radius of a circular arc R that lies on the sphere.

$$x^2 + y^2 = R^2$$

We can then find the mass $\rho\Delta V$ and position of a slice (Fig. 1.31).

$$\Delta m = \rho\pi x^2\Delta y \quad r(y) = y$$

Finally, we apply the continuous center of mass definition.

$$mr_{cm} = \int_0^R \rho\pi x^2 y dy = \frac{3m\pi}{2\pi R^3} * \int_0^R (R^2 - y^2) y dy$$

We can then expand the product and evaluate each part separately.

$$mr_{cm} = \frac{3m}{2R^3} * \left(\frac{R^2}{2}y^2 - \frac{1}{4}y^4 \right) \Big|_0^R$$

Solving for the center of mass, we find $r_{cm} = 3R/8$

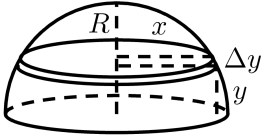


Figure 1.31

1.12 Trapezoid

Let the trapezoid have a mass m . We know the area $A = \frac{h}{2}(a+b)$ allowing us to solve for surface density $\sigma = m/A$.

We also know the width of the trapezoid decreases from $y = b$ to $y = a$ from $x = 0$ to $x = h$ meaning we can then create a line to model height as a function of distance.

$$y = \frac{a-b}{h}x + b$$

We can then find the mass $m = \sigma\Delta A$ and position of a slice (Fig. 1.32).

$$\Delta m = \sigma y\Delta x \quad r(x) = x$$

We finally apply the continuous center of mass definition.

$$mr_{cm} = \int_0^h \sigma y x dx = \frac{2m}{h(a+b)} * \int_0^h \left(\frac{a-b}{h}x + b \right) x dx$$

After some algebra, we find $r_{cm} = h(2a+b)/3(a+b)$

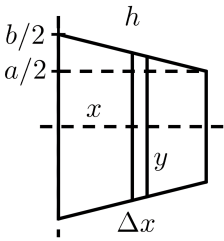


Figure 1.32

REMARKS: We can test a few limits to check our answers.

- $a = b$ results in $r_{cm} = h/2$, correctly telling us that the center of mass of a rectangle lies halfway in between.
- $a \gg b$ results in $r_{cm} \approx 2h/3$, correctly telling us that the center of mass of a triangle lies $2/3$ from the apex.
- $a \ll b$ results in $r_{cm} \approx h/3$, correctly telling us that the center of mass of a triangle lies $1/3$ from the base

1.13 Conical Frustum

Let the frustum have a mass m and density ρ . We will have to integrate to find the volume.

The radius increases from a to b over a distance h meaning we can model the radius as a function of distance. Then, we can find the mass $\rho\Delta V$ of a slice.

$$y = \frac{a-b}{h}x + b$$

$$\Delta m = \rho \pi y^2 \Delta x$$

We then add up the masses of all the slices to find the total mass (Fig. 1.33) .

$$m = \int_0^h \rho \pi y^2 dx = \rho \pi \int_0^h \left(\frac{a-b}{h}x + b \right)^2 dx$$

After some algebra, we find $m = \rho \pi h(a^2 + ab + b^2)/3$. We can now weight each slice with a position vector $r(x) = x$ and re-integrate to find the center of mass.

$$mr_{cm} = \int_0^h \rho \pi y^2 x dx = \frac{3m}{h(a^2 + ab + b^2)} \int_0^h \left(\frac{a-b}{h}x + b \right)^2 x dx$$

Letting a calculator do the work, we find $r_{cm} = h(3a^2 + 2ab + b^2)/4(a^2 + ab + b^2)$

REMARKS: We can check a few limits to test our answer.

- $a = b$ results in $r_{cm} = h/2$, correctly telling us that the center of mass of a cylinder lies halfway in between.
- $a \gg b$ results in $r_{cm} \approx 3h/4$, correctly telling us that the center of mass of a cone lies $3/4$ of the distance from the apex.
- $a \ll b$ results in $r_{cm} \approx h/4$, correctly telling us that the center of mass of a cone lies $1/4$ of the distance from the base.

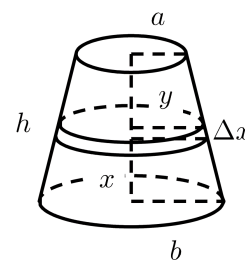


Figure 1.33

Chapter 2

Moment of Inertia

2.1 Definitions

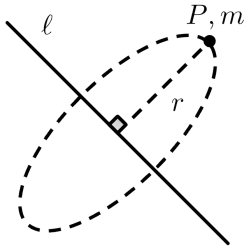


Figure 2.1

Point

Consider a line ℓ and a point P in space such that the point rotates in a circle around ℓ .

$$I = mr_{\perp}^2 \quad (2.1)$$

I represents the moment of inertia of P with respect to the axis of rotation ℓ ; m represents the mass of P ; r represents the perpendicular distance between P and ℓ . (Fig. 2.1)

Discrete Definition

Consider a collection of N points in space and a line ℓ that all the points rotate about.

$$I = \sum_{k=1}^N m_k r_k^2 \quad (2.2)$$

I represents the total moment of inertia of all N points; r_k represents the perpendicular distance between ℓ and the k th point; m_k represents the mass of the k th point.

Continuous Definition

Consider an object S and a line ℓ such that the S rotates about ℓ .

$$I = \int_S r^2 dm \quad (2.3)$$

I represents the moment of inertia of S ; dm represents an arbitrary differential mass; r represents the perpendicular distance between dm and ℓ . (Fig. 2.2)

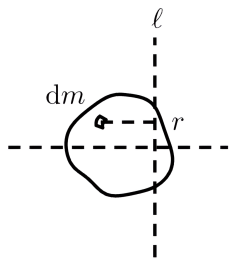


Figure 2.2

Superposition

The moment of inertia of an object equals the sum of the moment of inertias of its parts. If two parts are symmetrical with respect to the axis of rotation, they can be superimposed without changing the moment of inertia.

Equivalent Radius

An object S with mass m , major length d , and moment of inertia βmd^2 has the equivalent moment of inertia as that of a hoop with radius $r = d\sqrt{\beta}$.

Questions

Q2.1) Regardless of the origin O in a given coordinate system, why does only the perpendicular distance $r_{\perp}$ matter when determining moment of inertia?

Q2.2) Show how the discrete definition for the moment of inertia of a system follows from the definition of the moment of inertia of a point.

Q2.3) Show how the continuous definition for the moment of inertia of a system follows from the discrete definition of the moment of inertia of a system.

Q2.4) Make sure you intuitively understand how superposition follows from the associative property of addition when adding the moment of inertia of multiple points.

Q2.5) Show that if multiple point masses are symmetric with respect to the axis of rotation (all located a radial distance r away), they can be replaced by a single point with equivalent mass located at the same radial distance r .

Q2.6) We wish to extend the previous argument. Show that if multiple planar objects are symmetric with respect to the axis of rotation (all sharing the same major length ℓ and coefficient β), they can be replaced by a single planar object with equivalent mass.

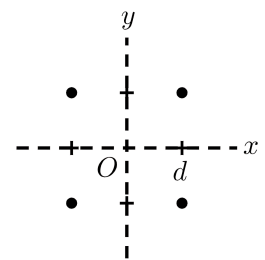
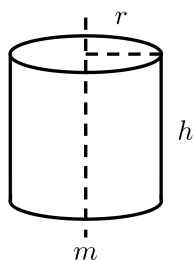
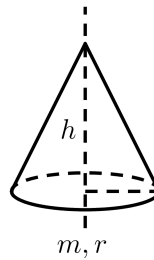
Q2.7) Confirm the prior statement by calculating the moment of inertia of the hoop $I = mr^2$ with radius $r = d\sqrt{\beta}$ and comparing it to the object S .

Exercises

E2.8) 4 points with mass m are located a radial distance $d\sqrt{2}$ from the origin O . Find the moment of inertia of the system with respect to the y -axis (Fig. 2.3).

E2.9) Find the moment of inertia of a hoop with mass m and radius r about its cylindrical axis.

E2.10) A cylinder has a mass m , radius r , and height h . Given a circle has a moment of inertia $I = \frac{1}{2}mr^2$, find the moment of inertia of the cylinder about its cylindrical axis (Fig. 2.4).

**Figure 2.3****Figure 2.4****Figure 2.5**

E2.11) The cylinder is now replaced with a hollow cone that also has a mass m and radius r . Find the new moment of inertia (Fig. 2.5).

E2.12) From $x = d$ to $x = Nd$, N equally sized points lie on the x -axis, with total mass m . Find the moment of inertia relative to a perpendicular axis.

2.2 Tools

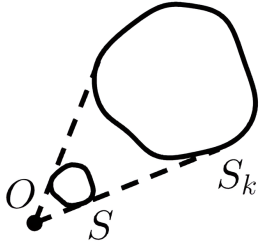


Figure 2.6

Scaling Arguments

Consider an object S that is scaled by a factor k about the axis of rotation O to form the object S_k .

$$I_k = I_0 k^{r+2} \quad (2.4)$$

I_k represents the moment of inertia of the scaled object S_k ; I_0 represents the original moment of inertia of the object S ; k represents the scale factor; r represents a integer constant that is dependent on the geometry of the object. (Fig. 2.6)

Limiting Arguments

Consider an object S_1 and a smaller scaled down version S_2 . Superimpose both objects such that the object S_3 is formed by their difference.

$$\lim_{S_2 \rightarrow S_1} I_3 = \lim_{S_2 \rightarrow S_1} I_1 - I_2 \quad (2.5)$$

I_1 , I_2 , and I_3 , represent the moment of inertia of S_1 , S_2 , and S_3 respectively. (Fig. 2.7)

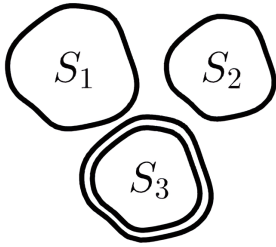


Figure 2.7

Questions

Q2.13) Recall what happens to mass when an object is scaled by a factor k . We seek to prove Eq. 2.4 by evaluating the change in the moment of inertia I of a point after scaling it by a factor k ; then, sum up the moment of inertias over the given system S .

Q2.14) Show that Eq. 2.5 follows from superposition and evaluate the limit.

2.3 Theorems

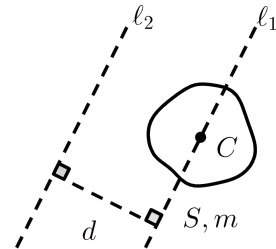


Figure 2.8

Parallel Axis Theorem

Consider an solid S in space with a line ℓ_1 that goes through its center of mass C and a line ℓ_2 that is parallel to ℓ_1 some distance away.

$$I = I_{cm} + md^2 \quad (2.6)$$

I represents the moment of inertia of S relative to ℓ_2 ; I_{cm} represents the moment of inertia of S relative to ℓ_1 ; m represents the total mass of S ; d represents the perpendicular distance between ℓ_1 and ℓ_2 . (Fig. 2.8)

Perpendicular Axis Theorem

Consider a shape S in the plane alongside the three perpendicular axes: x , y , and z , where z is perpendicular to the plane.

$$I_z = I_x + I_y \quad (2.7)$$

I_z represents the moment of inertia of S relative to the z axis; I_x and I_y represent the moment of inertias of S relative to the x and y axes. (Fig. 2.9)

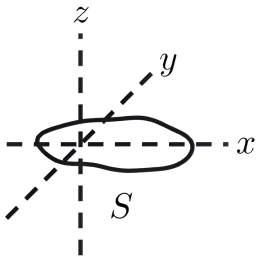


Figure 2.9

Three Axes Theorem

Consider a solid S in space alongside three perpendicular axes: x , y , and z , which are concurrent at some point P .

$$I_x + I_y + I_z = 2 \int_S r^2 dm \quad (2.8)$$

I_x , I_y , and I_z are the moment of inertias of the object relative to x , y , and z axes; dm represents a differential mass of S ; r represents the radial distance between an arbitrary differential mass dm and the point of concurrency P . (Fig. 2.10)

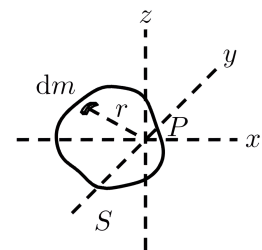


Figure 2.10

Questions

- Q2.15) Show that Eq. 2.6 reduces correctly for a point mass.
- Q2.16) Show that the minimum moment of inertia occurs when the axis of rotation ℓ crosses the center of mass.
- Q2.17) The moment of inertia of a rod with mass m and length ℓ when hinged at its center is given by $I = \frac{1}{12}m\ell^2$. Find its moment of inertia hinged at one end.
- Q2.18) A rod hinged at its center is merely the superposition of two rods hinged at their sides, yet they have different coefficients. Explain.
- Q2.19) Show that Eq. 2.7 reduces correctly for a point mass located on the x -axis, y -axis, and somewhere on the xy -plane but fails for a point located on the z -axis.
- Q2.20) Show that Eq. 2.8 reduces to Eq. 2.7 through evaluating I_z for a planar object.

Exercises

- E2.21) Find the moment of inertia of a hoop with mass m and radius r about its diametric axis.
- E2.22) A hoop has a mass m and radius r . It is then stretched by a factor a/r and b/r in the x and y directions respectively. Find the new moment of inertia about the cylindrical axis (Fig. 2.11)

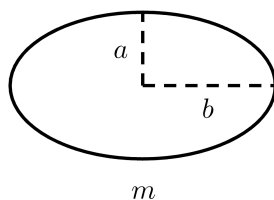


Figure 2.11

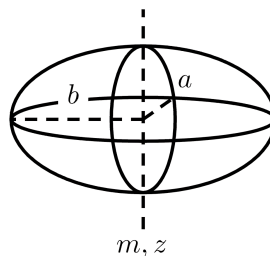


Figure 2.12

- E2.23) A hollow sphere ($I = \frac{2}{3}mr^2$) has a mass m and radius r . It is then stretched by a factor a/r and b/r in the x and y directions respectively. Find the new moment of inertia about the z -axis (Fig. 2.12).

2.4 Problems

2.1 Parallel Axis Theorem

A perpendicular line ℓ_1 goes through the center of mass of a planar blob S with mass m . Given the distance d between ℓ_2 , an axis parallel to ℓ_1 , derive Eq. 2.6.

2.2 Perpendicular Axis Theorem

Three axes x , y , and z are concurrent. Given that a blob S in the xy plane has a moment of inertia I_x , I_y , and I_z relative to each axis. Derive Eq. 2.7.

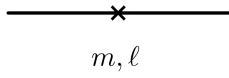


Figure 2.13

2.3 Three Axes Theorem

A blob S lies in space with a moment of inertia I_x , I_y , and I_z relative to the x , y , and z axes. Let r be the radial distance from the point of concurrency. Derive Eq. 2.8.

2.4 Rod

A line has a mass m and length ℓ . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.13) .

2.5 Square

A square has a mass m and side length ℓ . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.14).

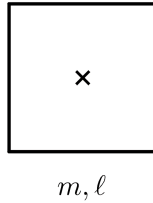


Figure 2.14

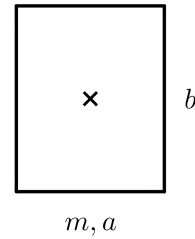


Figure 2.15

2.6 Rectangle

A rectangle has a mass m , height a , and width b . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.15).

2.7 Equilateral Triangle

An equilateral triangle has a mass m and side length ℓ . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.16).

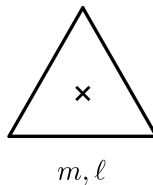


Figure 2.16

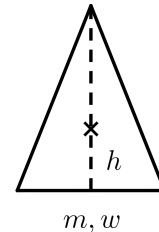


Figure 2.17

2.8 Isosceles Triangle

An isosceles triangle has a mass m , width w , and height h . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.17).

2.9 Circle

A circle has a mass m and radius r . Find the moment of inertia relative to a perpendicular axis located at the center of mass (Fig. 2.18) .

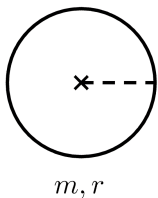


Figure 2.18

2.10 Cylinder

A cylinder has a mass m , radius r , and height h . Find the moment of inertia relative to axis through the center of mass, parallel to each circular face (Fig. 2.19).

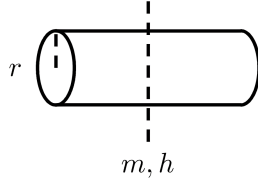


Figure 2.19

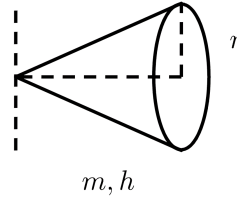


Figure 2.20

2.11 Solid Cone

A solid cone has a mass m , height h , and radius r . Find the moment of inertia relative to an axis through the apex and parallel to the base (Fig. 2.20).

2.12 Spherical Shell

A spherical shell has a mass m and radius r . Find the moment of inertia relative to an axis through the center of mass (Fig. 2.21).

2.13 Solid Sphere

A solid sphere has a mass m and radius r . Find the moment of inertia relative to an axis through the center of mass.

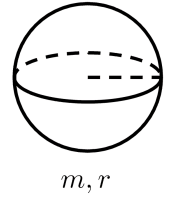


Figure 2.21

2.5 Calculus Solutions to Problems

2.1 Parallel Axis Theorem

We can express the position $\mathbf{r}'$ of a point P as the sum of the position of the center of mass $\mathbf{r}_{cm}$ and the point's position relative to the center of mass $\mathbf{r}$ (Fig. 2.22).

$$\mathbf{r}' = \mathbf{r}_{cm} + \mathbf{r}$$

The dot product of any vector with itself equals the magnitude squared $\mathbf{r} \cdot \mathbf{r} = r^2$. Using this, we can rewrite the continuous moment of inertia definition.

$$I = \int_S \mathbf{r}' \cdot \mathbf{r}' dm = \int_S (\mathbf{r}_{cm} + \mathbf{r}) \cdot (\mathbf{r}_{cm} + \mathbf{r}) dm$$

We then recall the definition of total mass, the value of total position-mass in the center of mass origin, and the moment of inertia about the center of mass.

$$m = \int_S dm \quad 0 = \int_S \mathbf{r} dm \quad I_{cm} = \int_S r^2 dm$$

Letting the center of mass have a magnitude $|\mathbf{r}_{cm}| = d$, we carry out all the algebra making the corresponding substitutions.

$$\begin{aligned} I &= \int_S (\mathbf{r}_{cm} \cdot \mathbf{r}_{cm} + 2\mathbf{r}_{cm} \cdot \mathbf{r} + \mathbf{r} \cdot \mathbf{r}) dm \\ &= d^2 \int_S dm + 2\mathbf{r}_{cm} \cdot \int_S \mathbf{r} dm + \int_S r^2 dm \end{aligned}$$

We see that the center of mass about a parallel axis is $I = I_{cm} + md^2$

REMARK: If the two axes weren't parallel, our vector addition substitution wouldn't be possible. If the second axis didn't pass through the center of mass, the last two integrals wouldn't reduce correctly. We can generalize this result to a 3D-blob using squashing.

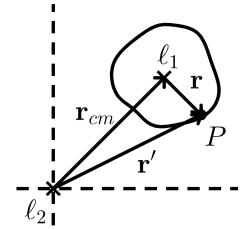


Figure 2.22

2.2 Perpendicular Axis Theorem

Using the Pythagorean Theorem, we can express the radial distance r of a point P from the z axis in terms of the components x and y (Fig. 2.23) .

$$r^2 = x^2 + y^2$$

We then recall the continuous moment of inertia definition relative to each of the three axes.

$$I_x = \int_S y^2 dm \quad I_y = \int_S x^2 dm \quad I_z = \int_S r^2 dm$$

Finally, we carry out the algebra starting from the Pythagorean Theorem.

$$\begin{aligned} r^2 dm &= (x^2 + y^2) dm \\ \int_S r^2 dm &= \int_S x^2 dm + \int_S y^2 dm \end{aligned}$$

Substituting our definitions, we see $I_z = I_x + I_y$

REMARK: If the object isn't planar, the distance from the x -axis would be r_x , not y , as the height of the point (the z position) would affect the radial distance.

2.3 Three Axes Theorem

Let r_z , r_y , and r_x represent the perpendicular distances away from the z , y , and x axes. Then, let r represent the radial distance away from the origin (Fig. 2.24) .

Again, we can use the Pythagorean theorem to relate each perpendicular distance to 2 components.

$$r_x^2 = y^2 + z^2 \quad r_y^2 = z^2 + x^2 \quad r_z^2 = x^2 + y^2$$

Then, we can reuse the Pythagorean theorem to relate the components to the net radial distance.

$$r^2 = x^2 + y^2 + z^2$$

Finally, we recall the continuous moment of inertia definition relative to each of the three axes.

$$I_x = \int_S r_x^2 dm \quad I_y = \int_S r_y^2 dm \quad I_z = \int_S r_z^2 dm$$

Finally, we carry out the algebra starting from the Pythagorean Theorem.

$$\begin{aligned} r_x^2 + r_y^2 + r_z^2 &= 2(x^2 + y^2 + z^2) = 2r^2 \\ \int_S r_x^2 dm + \int_S r_y^2 dm + \int_S r_z^2 dm &= 2 \int_S r^2 dm \end{aligned}$$

Substituting our definitions, we see $I_x + I_y + I_z = 2 \int_S r^2 dm$

REMARK: In the diagram, dashed lines represent the perpendicular distances; solid lines represent the components of the position vector; the vector represents the radial distance from the origin. Try to spot all the triangles we applied the Pythagorean theorem to.

2.4 Rod

The rod has a linear density $\lambda = m/\ell$ allowing us to find the mass and perpendicular distance of a slice (Fig. 2.25) .

$$\Delta m = \lambda \Delta x \quad r_{\perp}^2 = x^2$$

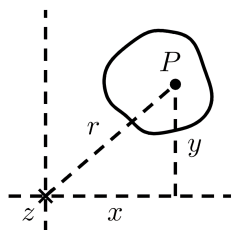


Figure 2.23

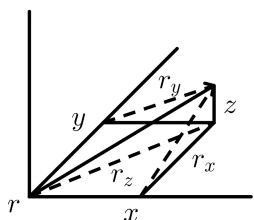


Figure 2.24

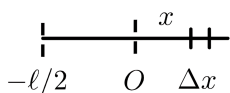


Figure 2.25

We then apply the continuous moment of inertia definition.

$$I = \int_{-\ell/2}^{\ell/2} \lambda x^2 dx = \frac{m}{\ell} * \frac{1}{3} x^3 \Big|_{-\ell/2}^{\ell/2}$$

Evaluating the expression, we find $I = \frac{1}{12} m \ell^2$

2.5 Square

The square has a surface density $\sigma = m/\ell^2$ allowing us to find the mass $\sigma \Delta A$ and the perpendicular distance of a slice (Fig. 2.26) .

$$\Delta m = \sigma \Delta x \Delta y \quad r_{\perp}^2 = x^2 + y^2$$

We then apply the continuous moment of inertia definition.

$$I = \int_{-\ell/2}^{\ell/2} \int_{-\ell/2}^{\ell/2} \sigma (x^2 + y^2) dx dy = \frac{m}{\ell^2} \left[\left(\int_{-\ell/2}^{\ell/2} x^2 dx \right) \ell + \ell \left(\int_{-\ell/2}^{\ell/2} y^2 dy \right) \right]$$

We find the moment of inertia to be $I = \frac{1}{6} m \ell^2$

REMARK: To evaluate the integral, we broke the integral into two parts: one term for x^2 and one term for y^2 . In each, the other variable dy and dx is constant, allowing it to be factored out. We could then easily evaluate all the component integrals. If you are uncomfortable with the properties of double summation, you can instead cut the square into rods and apply the parallel axis theorem to each rod using $dI = \frac{1}{12} \ell^2 dm + x^2 dm$.

2.6 Rectangle

We could repeat the exact same process as that of a square. The only difference will be the surface density $\sigma = m/ab$ and the limits of integration.

Instead, we'll find the moment of inertia of one quarter of the rectangle, and multiply by four. Our resulting integral will be:

$$I = 4 \int_0^{a/2} \int_0^{b/2} \sigma (x^2 + y^2) dx dy$$

Evaluating the integral in the same process as before, we find $I = \frac{m}{12} (a^2 + b^2)$.

REMARK: A rectangle can be constructed from 4 smaller rectangles hinged at their ends. We see that superposition / symmetry can simplify our integrals. Additionally, we can check the limit $a = b \equiv \ell$. The moment of inertia correctly reduces to $I = \frac{1}{6} m \ell^2$, that of a square.

2.7 Equilateral Triangle

We can simplify our algebra find finding the moment of inertia relative to one corner I' and utilizing the Parallel Axis Theorem in reverse to solve for the center of mass moment of inertia.

The equilateral triangle has an area $A = \frac{\sqrt{3}}{4} \ell^2$ and a surface density m/A . We then cut the triangle into rods, finding the mass $\sigma \Delta A$ and moment of inertia $md^2 + \frac{1}{12} m \ell^2$ of each slice using the parallel axis theorem.

$$\Delta m = \sigma (2y) \Delta x \quad \Delta I = x^2 \Delta m + \frac{1}{12} (2y)^2 \Delta m$$

The interior angles of an equilateral triangle are $\pi/3$, so the half angle is $\pi/6$. Using trigonometry (Fig. 2.27) :

$$\tan \frac{\pi}{6} = \frac{y}{x} = \frac{1}{\sqrt{3}}$$

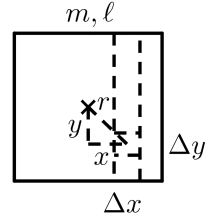


Figure 2.26

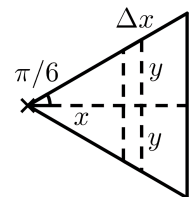


Figure 2.27

Finally, we apply $I = \int dI$, an alternate form of the continuous moment of inertia definition using superposition.

$$I' = \int_0^{\sqrt{3}\ell/2} \sigma(2y)(x^2 + y^2/3) dx = \frac{2 * 4m}{\sqrt{3}\ell^2} \int_0^{\sqrt{3}\ell/2} (x/\sqrt{3})(x^2 + x^2/9) dx$$

We can then simplify even further and evaluate the integral.

$$I' = \frac{2^3 m}{3\ell^2} \int_0^{\sqrt{3}\ell/2} \frac{10}{9} x^3 dx = \frac{2^4 * 5m}{3^3 \ell^2} * \frac{1}{4} x^4 \Big|_0^{\sqrt{3}\ell/2}$$

We find $I' = \frac{5}{12} m\ell^2$. We then apply the Parallel Axis Theorem $I' = md^2 + I_{cm}$ with $d = \ell/\sqrt{3}$.

$$\frac{5}{12} m\ell = m(\ell/\sqrt{3})^2 + I_{cm}$$

Solving, we find $I_{cm} = \frac{1}{12} m\ell^2$

REMARKS: Integrating around the center of mass can get pretty messy. We could've cut the equilateral triangle into 6 right triangles. Once we found the moment of inertia of one, we could find the total by multiplying by 6. However, we see that a little cleverness goes a far way in simplifying our algebra.

2.8 Isosceles Triangle

We apply a similar strategy as before. The isosceles triangle has an area $A = wh/2$ and a surface density $\sigma = m/A$. Cutting the triangle into rods, we find the mass $\sigma\Delta A$ and moment of inertia $md^2 + \frac{1}{12} m\ell^2$ of each.

$$\Delta m = \sigma(2y)\Delta x \qquad \Delta I = x^2 \Delta m + \frac{1}{12} (2y)^2 \Delta m$$

We can then use similar triangles to relate x and y (Fig. 2.28) .

$$\frac{x}{y} = \frac{h}{w/2}$$

Finally, we apply $I = \int dI$ to solve for the total moment of inertia.

$$I' = \int_0^h \sigma(2y)(x^2 + y^2/3) dx = \frac{2 * 2m}{wh} \int_0^h \left(\frac{w}{2h}x\right) \left(x^2 + \frac{w^2}{3 * 4h^2}x^2\right) dx$$

Simplifying, we find

$$I' = \frac{2m}{h^2} * \frac{12h^2 + w^2}{12h^2} \int_0^h x^3 dx = \frac{m(12h^2 + w^2)}{6h^4} * \frac{1}{4} x^4 \Big|_0^h$$

Solving, we find $I' = \frac{1}{2} mh^2 + \frac{1}{24} mw^2$. Applying the Parallel Axis Theorem $I' = md^2 + I_{cm}$, we find

$$\frac{1}{2} mh^2 + \frac{1}{24} mhw = m(2h/3)^2 + I_{cm}$$

Solving, we see $I_{cm} = \frac{1}{18} mh^2 + \frac{1}{24} mw^2$

REMARKS: Naturally, we should test our answer with the limits $(h, w) = (\frac{\sqrt{3}}{2}\ell, \ell)$ for an equilateral triangle. Testing I_{cm} , we find $I_{cm} = \frac{1}{12} m\ell^2$; testing I' , we find $I' = \frac{5}{12} m\ell^2$. Both results agree with those of our equilateral triangle.

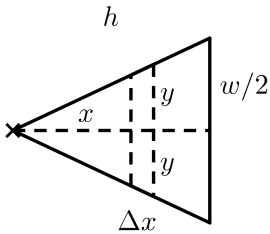


Figure 2.28

2.9 Circle

The circle has a surface density $\sigma = m/\pi r^2$. We can cut the circle into hoop-like slices. We find the mass $\sigma \Delta A$ and perpendicular distance of a slice (Fig. 2.29).

$$\Delta m = 2\pi x \Delta x \quad r_{\perp}^2 = x^2$$

We then apply the continuous moment of inertia definition.

$$I = \int_0^r \sigma 2\pi x^3 dx = \frac{m * 2\pi}{\pi r^2} * \frac{1}{4} x^4 \Big|_0^r$$

Evaluating, we find $I = \frac{1}{2} m r^2$

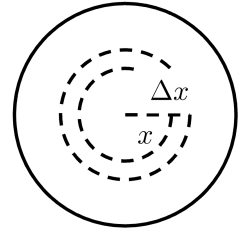


Figure 2.29

2.10 Cylinder

The cylinder has a linear density $\lambda = m/h$. We can cut the cylinder into disks, applying the parallel axis theorem to each slice.

Recall that the moment of inertia of a disk about a diametric axis is given by $I_d = mr^2/4$. We then find the mass and moment of inertia of each slice.

$$\Delta m = \lambda \Delta x \quad \Delta I = \Delta m x^2 + \Delta m r^2 / 4$$

Then, we apply the continuous moment of inertia definition.

$$I = \int_{-\ell/2}^{\ell/2} \lambda (x^2 + r^2/4) dx = \frac{m}{\ell} \left(\frac{1}{3} x^3 + \frac{r^2}{4} x \right) \Big|_{-\ell/2}^{\ell/2}$$

Solving, we see $I = \frac{1}{12} m \ell^2 + \frac{1}{4} m r^2$

REMARKS: We should test some limits.

- $r \ll \ell$ results in $I \approx \frac{1}{12} m \ell^2$; the cylinder correctly becomes a rod for small r .
- $r \gg \ell$ results in $I \approx \frac{1}{4} m r^2$; the cylinder correctly becomes a disk for large r .

2.11 Solid Cone

The cone has a volume $V = \pi r^2 h/3$ and volume density $\rho = m/V$. We can cut the cone into disks as before and apply the parallel axis theorem to each slice.

Using similar triangles, we see

$$\frac{y}{x} = \frac{r}{h}$$

Using $I_d = \frac{1}{4} m r^2$, we can find the mass $\rho \Delta V$ and moment of inertia of each slice (Fig. 2.30).

$$\Delta m = \pi y^2 \Delta x \quad \Delta I = \Delta m x^2 + \Delta m y^2 / 4$$

Then, we apply the continuous center of mass definition.

$$I = \int_0^h \rho \pi y^2 (x^2 + y^2/4) dx = \frac{3m * \pi}{\pi r^2 h} \int_0^h \frac{r^2}{h^2} x^2 \left(x^2 + \frac{r^2}{4h^2} x^2 \right) dx$$

We can then simplify the integral.

$$I = \frac{3m}{h^3} \left(\frac{4h^2 + r^2}{4h^2} \right) \int_0^h x^4 dx = \frac{3m(4h^2 + r^2)}{4h^5} * \frac{1}{5} x^5 \Big|_0^h$$

Finally, we evaluate to find $I = \frac{3}{5} m h^2 + \frac{3}{20} m r^2$

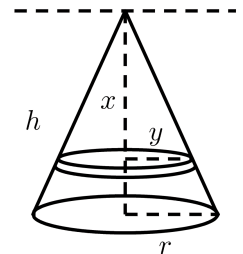


Figure 2.30

REMARK: A rod with mass m , density $\rho = x^n$, and length ℓ will have a moment of inertia $\frac{n}{n+2} m \ell^2$. Take a second to confirm this by either integrating or plugging in $n = 1$ for a uniform density rod which we have already derived. For a cone with a small radius $r \ll h$, it effectively acts as a rod with $n = 2$. Correctly, a small radius results in $I \approx \frac{3}{5} m h^2$.

2.12 Spherical Shell

The spherical shell has a surface density $\sigma = m/4\pi r^2$. We can find the mass $\sigma\Delta A$ and the perpendicular distance of each slice (Fig. 2.31).

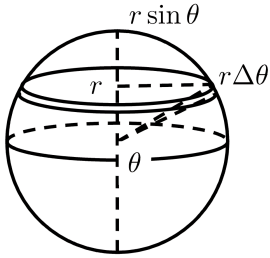


Figure 2.31

$$\Delta m = \sigma 2\pi r \sin \theta * r \Delta \theta$$

$$r_{\perp}^2 = r^2 \sin^2 \theta$$

We then apply the continuous moment of inertia definition alongside the trigonometric identity $\sin^2 \theta + \cos^2 \theta = 1$

$$I = \int_0^\pi 2\pi \sigma r^4 \sin^3 \theta d\theta = \frac{2\pi r^4 * m}{4\pi r^2} \int_0^\pi \sin \theta (1 - \cos^2 \theta) d\theta$$

Next, we can break the integral into two parts making the substitution $u = \cos \theta$ and $du = -\sin \theta d\theta$

$$I = \frac{mr^2}{2} \left(\int_0^\pi \sin \theta d\theta + \int_1^{-1} u^2 du \right) = \frac{mr^2}{2} \left(-\cos \theta \Big|_0^\pi + \frac{1}{3} u^3 \Big|_1^{-1} \right)$$

Evaluating the expression, we find $I = \frac{2}{3}mr^2$

REMARK: We see that the moment of inertia is more than a disk $I = \frac{1}{2}mr^2$ and less than a hoop $I = mr^2$. That's because if you were to squash a hollow sphere, you would get a disk with greater density on the outside (think about why the curvature causes this).

2.13 Solid Sphere

A sphere has a volume $V = \frac{4}{3}\pi r^3$ and a volume density $\rho = m/V$. Using Pythagorean's theorem, we can then relate the radius of a slice x , height y , and the radius of a circular arc r that lies on the sphere (Fig. 2.32).

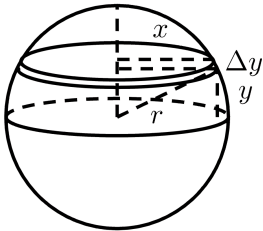


Figure 2.32

$$x^2 + y^2 = r^2$$

Next, we can find the mass $\rho\Delta V$ and moment of inertia of a disk-like slice.

$$\Delta m = \rho \pi x^2 \Delta y$$

$$\Delta I = \Delta m x^2 / 2$$

Finally, we apply the continuous moment of inertia definition.

$$I = \int_{-r}^r \frac{\rho \pi}{2} x^4 dy = \frac{3m * \pi}{4\pi r^3 * 2} \int_{-r}^r (r^2 - y^2)^2 dy$$

Expanding and evaluating each polynomial individually, we find $I = \frac{2}{5}mr^2$

REMARK: We see that the moment of inertia is less than a disk $I = \frac{1}{2}mr^2$. If we were to squash the solid sphere, the density would be greater towards the center, meaning the mass is closer on average to the axis of rotation.